

MA10509: Mathematics for Data Science
Practice Questions
Unit – 5: Data Science Tools and Case Studies

Short Answer Questions:

1. What are data tools and why are they used?
2. List any two benefits of using data tools.
3. Mention two applications of data tools in real-world industries.
4. Write the shortcut keys for Cut, Copy, and Paste in Excel.
5. Name any two types of charts available in MS Excel.
6. Who developed the R programming language and where was it developed?
7. Name any two features of R programming.
8. Write any two examples of basic arithmetic operations in R.
9. What is MATLAB and what does it stand for?
10. Why MATLAB is considered suitable for mathematical and engineering applications?

Long Answer Questions:

1. What are the key features, applications, challenges, and future trends of data tools?
2. Discuss how modern requirements such as scalability, security, and real-time processing shape the evolution of data tools.
3. Explain the steps involved in creating, saving, opening, and printing a workbook in MS Excel. Include the purpose of each step using example.
4. Discuss various types of charts such as Column, Line, Pie, Bar, and Area charts. Explain steps to create a chart with an example.
5. Explain the features of R programming language in detail. Discuss why R is widely used in data science and research fields.
6. Describe the different data structures available in R with suitable examples. Explain how vectors, matrices, lists, and data frames differ in terms of structure and usage.
7. Explain the key features of MATLAB in detail.
8. Discuss the major applications, advantages, and limitations of MATLAB. Explain how MATLAB is used in areas such as numerical computation, data visualization and artificial intelligence.